**Thermodynamics and Statistical Physics**

2nd Exercise Class

*Thermodynamic States, Reversible Processes, Work*

# Exercises for the Class

1. Consider a gas contained in the cylinder-piston arrangement shown in Fig. 1. The gas is pure nitrogen ($N_2$) with a mass of $m = 1.60$ g and is in equilibrium with its surroundings. The gas is initially at standard temperature and pressure, with $p = 101$ kPa and $T = 0$ °C. (The molar mass of nitrogen gas is $M = 28$ g/mol.)

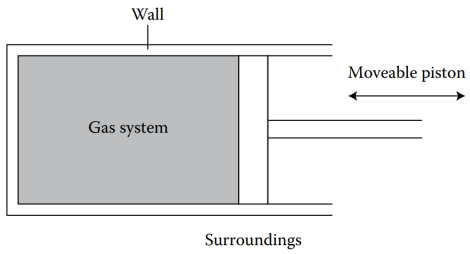



Fig. 1

   (a) Find the gas's volume.
   (b) The piston is now moved, slowly enough so that thermal equilibrium with the surroundings is maintained, until its pressure reaches 120 kPa. What is the gas's new volume?

2. An ideal gas undergoes the following reversible cycle: (i) an isobaric expansion from the state $(p_1, V_1)$ to the state $(p_1, V_2)$; (ii) an isochoric reduction in pressure to the state $(p_2, V_2)$; (iii) an isobaric reduction in volume to the state $(p_2, V_1)$; (iv) an isochoric increase in pressure back to the original state $(p_1, V_1)$. How much work is done on the gas in traversing the cycle 100 times if $p_1 = 3.0$ atm, $p_2 = 1.0$ atm, $V_1 = 1.0$ L and $V_2 = 2.0$ L?

3. 2.0 moles of an ideal gas is initially at $p =$ $= 1.0$ atm and $T = 300$ K. It is then taken through a three-step reversible process: (i) isobaric expansion to twice its original volume; (ii) isothermal compression, returning to the original volume; (iii) isochoric reduction in pressure to the original state. Find the work done on the gas in each step of the process and the net work done on the gas for the process.

4. One mole of an ideal gas is warmed slowly so that it goes from the state $(p_1, V_1)$ to $(3p_1, 3V_1)$ in such a way that the pressure of the gas is directly proportional to the volume.
   (a) How much work is done on the gas in the process?
   (b) How is the temperature of the gas related to its volume during this process?

5. Ideal gas performs the process shown in Fig. 2, $p_0 = 10^5$ Pa, $V_0 = 1$ dm$^3$. What is the volume associated with the maximum temperature state?

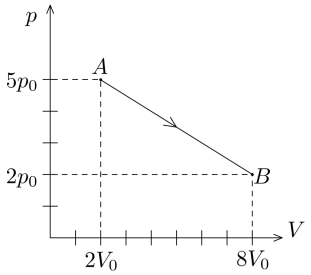



Fig. 2

6. The thin tube of cross-sectional area $A$ shown in the Fig. 3, soldered at one end, is located in a vertical plane. A mercury column of length $L$ in the horizontal part of the tube, length $3L$, encloses oxygen gas of length $L$. The external pressure $p_0$ is equal to the hydrostatic pressure of the mercury column of height $L$. The tube is in an environment whose temperature changes in such a way that the volume of the oxygen gas triples slowly. How much work did the expanding gas do?

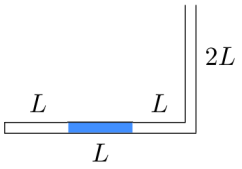



Fig. 3

7. A hypothetical substance has an isothermal compressibility $\kappa = a/v$ and volume expansion coefficient $\beta = 2bT/v$, where $a$ and $b$ are constants and $v$ is the molar volume. Show that the equation of state is

$$v - bT^2 + ap = \text{constant}.$$

8. The equation of state of a rubber band is

$$F = aT\left[\frac{L}{L_0} - \left(\frac{L_0}{L}\right)^2\right]$$

where $L_0$ is the original length and $a$ is a constant equal to $1.3 \cdot 10^{-2}$ N/K. How much work is performed when the band is stretched isothermally and reversibly from its original length of 10 cm to a length of 20 cm at a temperature of 20 °C?

## Homework to Be Submitted

1. In the $p$-$V$ diagram shown in Fig. 4, we plotted the isotherms corresponding to temperatures $T_1$ and $T_2$ for $n$ moles of gas, and drew a line $AC$ parallel to the $V$ axis connecting the two isotherms. Then, we marked points $B$ and $D$ cut out from the isotherms by the line which perpendicularly bisects segment $AC$.

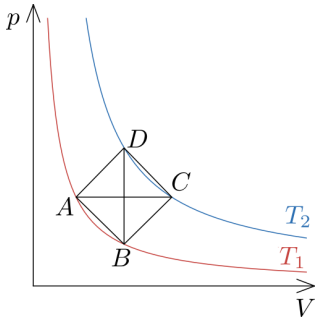



Fig. 4

   (a) Using gas laws, prove that ABCD is a rhombus.

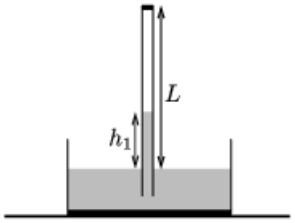
   (b) Let us assume that the gas undergoes the ADCBA cycle. Express the net work gained during one cycle with $n, T_1$, and $T_2$.

## Homework for Practice

1. During a reversible adiabatic expansion of an ideal gas, the pressure and volume at any moment are related by $pV^\gamma = c$ where $c$ and $\gamma$ are constants. Show that the work done by the gas in expanding from a state $(p_1, V_1)$ to a state $(p_2, V_2)$ is

$$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1}.$$

2. Some argon gas is put into a Torricelli-tube of length $L = 1$ m and of cross section $A = 2$ cm$^2$, therefore the height of the mercury in the tube is only $h_1 = 0.4$ m. The ambient air pressure is $p_0 = 10^5$ Pa, and the initial temperature is 20 °C.

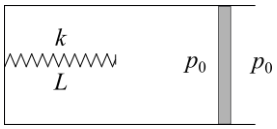



   (a) What is the mass of the argon gas above the mercury?

   (b) The temperature of the gas is slowly increased. What is the temperature of the gas when the height of the mercury in the tube is $h_2 = 0.36$ m?

   (c) How much work was done by the extending gas during the process?

3. A piston in a cylinder with a closed end and a cross-sectional area of $A = 0.5$ dm$^2$ traps air at a pressure of $p_0 = 10^5$ Pa and a volume of $V_0 = 4$ dm$^3$. A horizontal, upstretched spring with a length of $L = 4$ dm and a spring constant of $k = 625$ N/m is attached to the closed end of the cylinder. We change the temperature of the air in the cylinder so that during the very slow movement of the piston, the volume of the enclosed air decreases to $V_1 = 1$ dm$^3$.




   (a) How many times greater is the absolute temperature of the air in the initial state than in the final state?

   (b) Plot the pressure of the air in the cylinder as a function of its volume.

   (c) How much work was done when the air was compressed?

4. Given the state equation $p = p(T, V)$, express the quantity

$$\left(\frac{\partial p}{\partial T}\right)_V$$

using the thermal expansion coefficient $\beta$ and the isothermal compressibility $\kappa$.